



PROTECCIONES ELÉCTRICAS

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Sesión 2

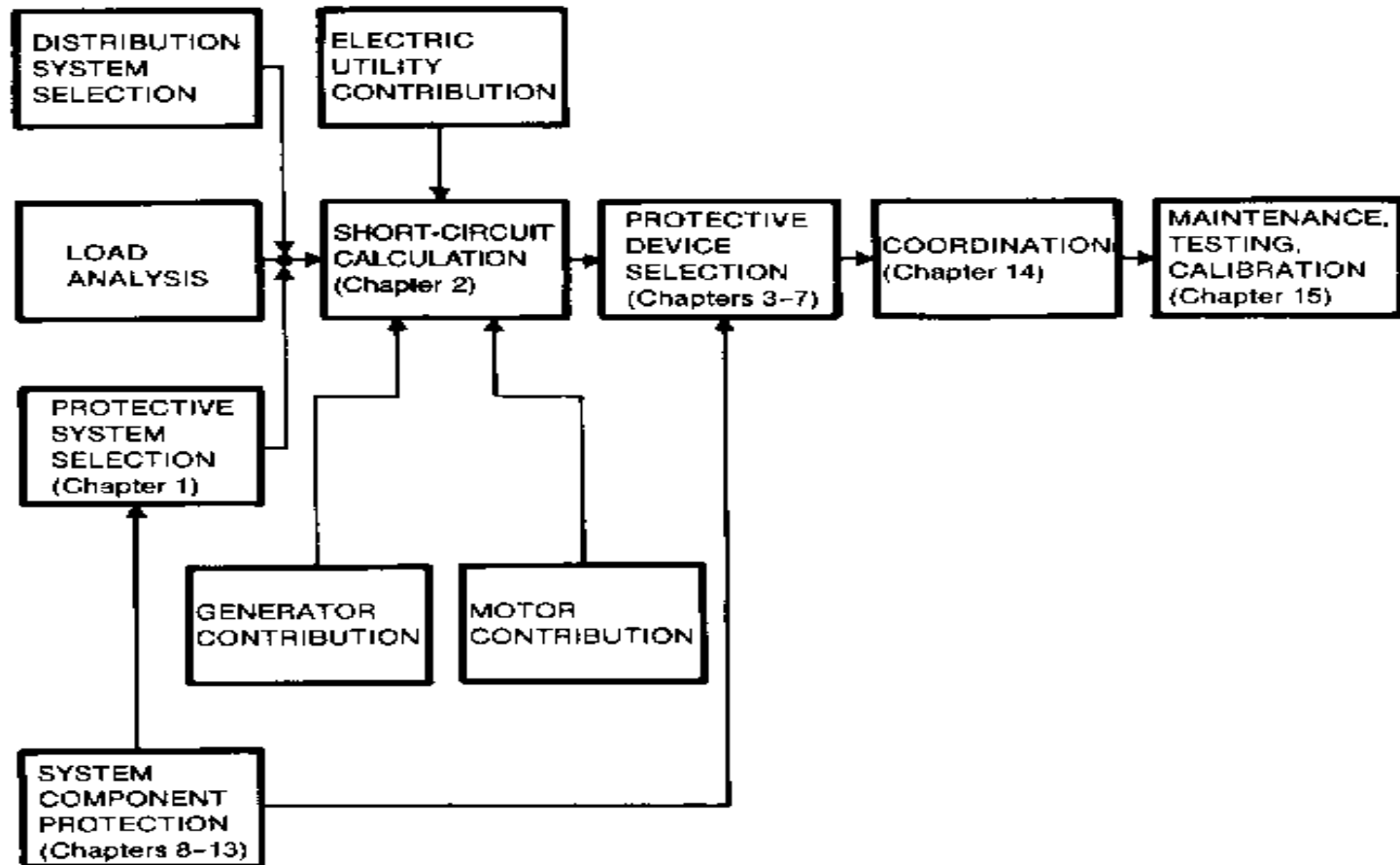
Cálculos de cortocircuito

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OBJETO DE LAS PROTECCIONES

Detectar y limitar la duración de una falla en
el sistema eléctrico de potencia

SECUENCIA DE PASOS EN UN SISTEMA DE PROTECCIONES ELÉCTRICAS. (IEEE STD 242)



ANÁLISIS DE FLUJO DE CARGA

TIPOS DE NODO	VALORES QUE SE DEFINEN	INCÓGNITAS
COMPENSADOR	$V, \delta = 0$	P, Q
GENERADOR	V, P	δ, Q
CARGA	P, Q	V, δ

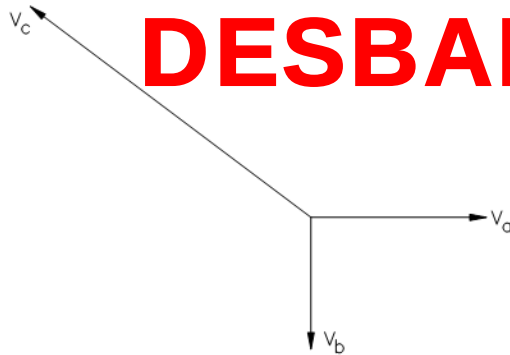
ANÁLISIS DE FLUJO DE CARGA

- ❏ Determinación de tensiones en los diferentes barrajes del sistema, para las condiciones de operación.
 - ❏ Determinación de las condiciones de cargabilidad de los elementos del sistema
 - ❏ Entrega tensiones y ángulos con los cuales se calculan I , P , Q , $\cos\phi$ y pérdidas
 - ❏ Flujos de potencias activa y reactiva entre nodos
-

PROCEDIMIENTO CÁLCULO CORTOCIRCUITO

- 1. Elaborar Diagrama Unifilar del sistema eléctrico de potencia.*
 - 2. Calcular Impedancias de los equipos: líneas, generadores, reactores, bancos de condensadores, transformadores y motores.*
 - 3. Elaborar diagrama de impedancias en P.U.*
 - 4. Normas: ANSI: C37.10, C37.13*
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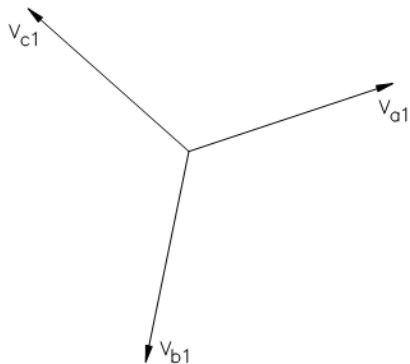
COMPONENTES SIMÉTRICAS DE UN SISTEMA TRIFÁSICO DESBALANCEADO



$$V_a = 8.0 \angle 0^\circ$$

$$V_b = 6.0 \angle -90^\circ$$

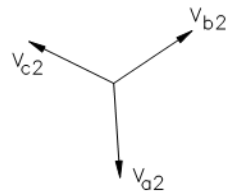
$$V_c = 16.0 \angle 143.1^\circ$$



$$V_{a1} = 9.8 \angle 18.4^\circ$$

$$V_{b1} = 9.8 \angle -101.6^\circ$$

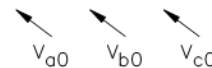
$$V_{c1} = 9.8 \angle 138.4^\circ$$



$$V_{a2} = 4.3 \angle -86.2^\circ$$

$$V_{b2} = 4.3 \angle 33.8^\circ$$

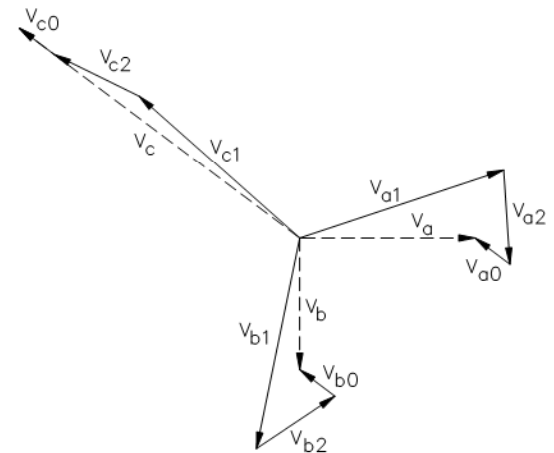
$$V_{c2} = 4.3 \angle -206.2^\circ$$



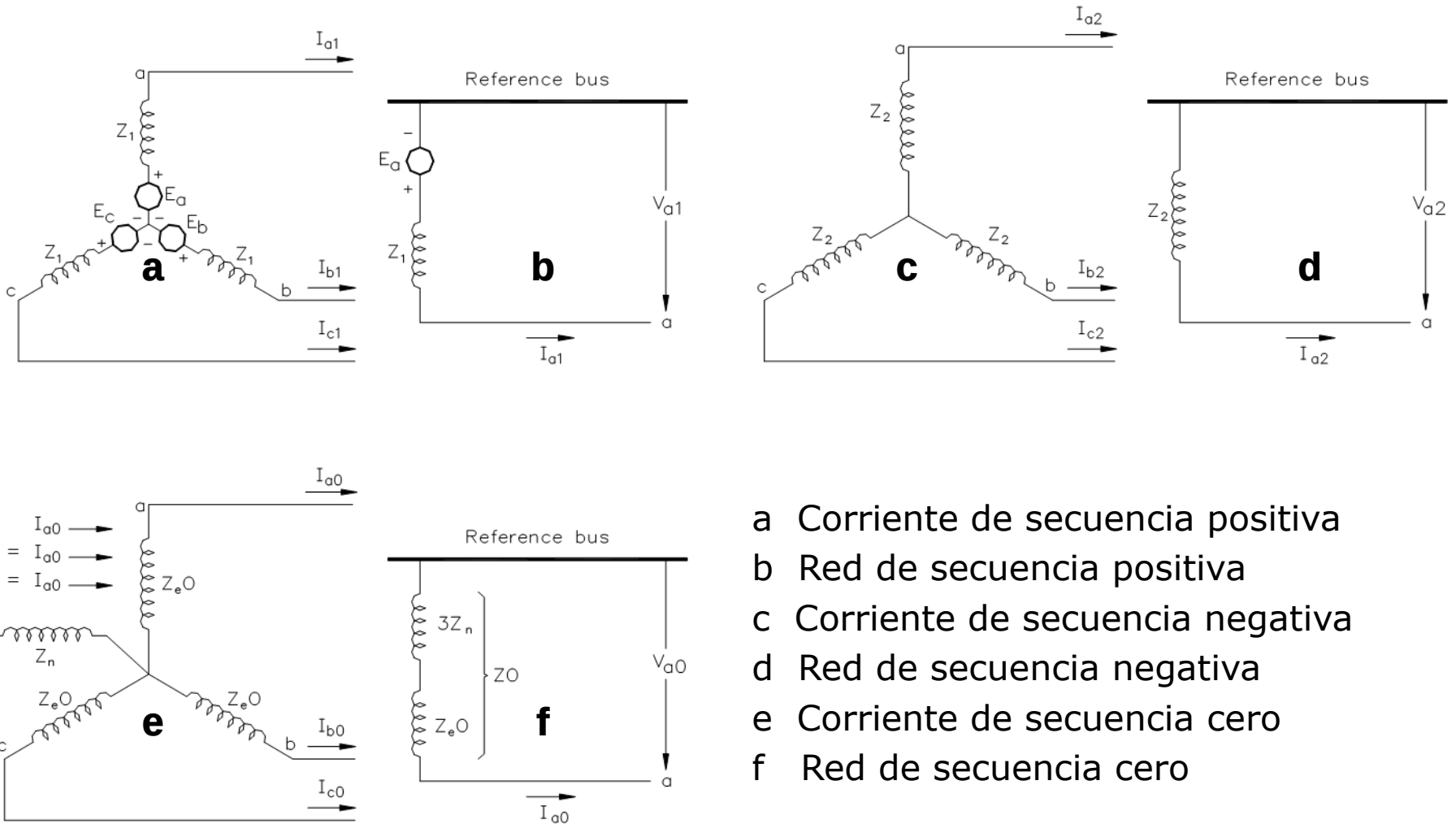
$$V_{a0} = 2.0 \angle 143.1^\circ$$

$$V_{b0} = 2.0 \angle 143.1^\circ$$

$$V_{c0} = 2.0 \angle 143.1^\circ$$



REDES DE SECUENCIA



IMPEDANCIAS DE SECUENCIA

LÍNEAS:

$$Z_1 = Z_2; \quad Z_0 = (2 \text{ a } 3.5)Z_1$$

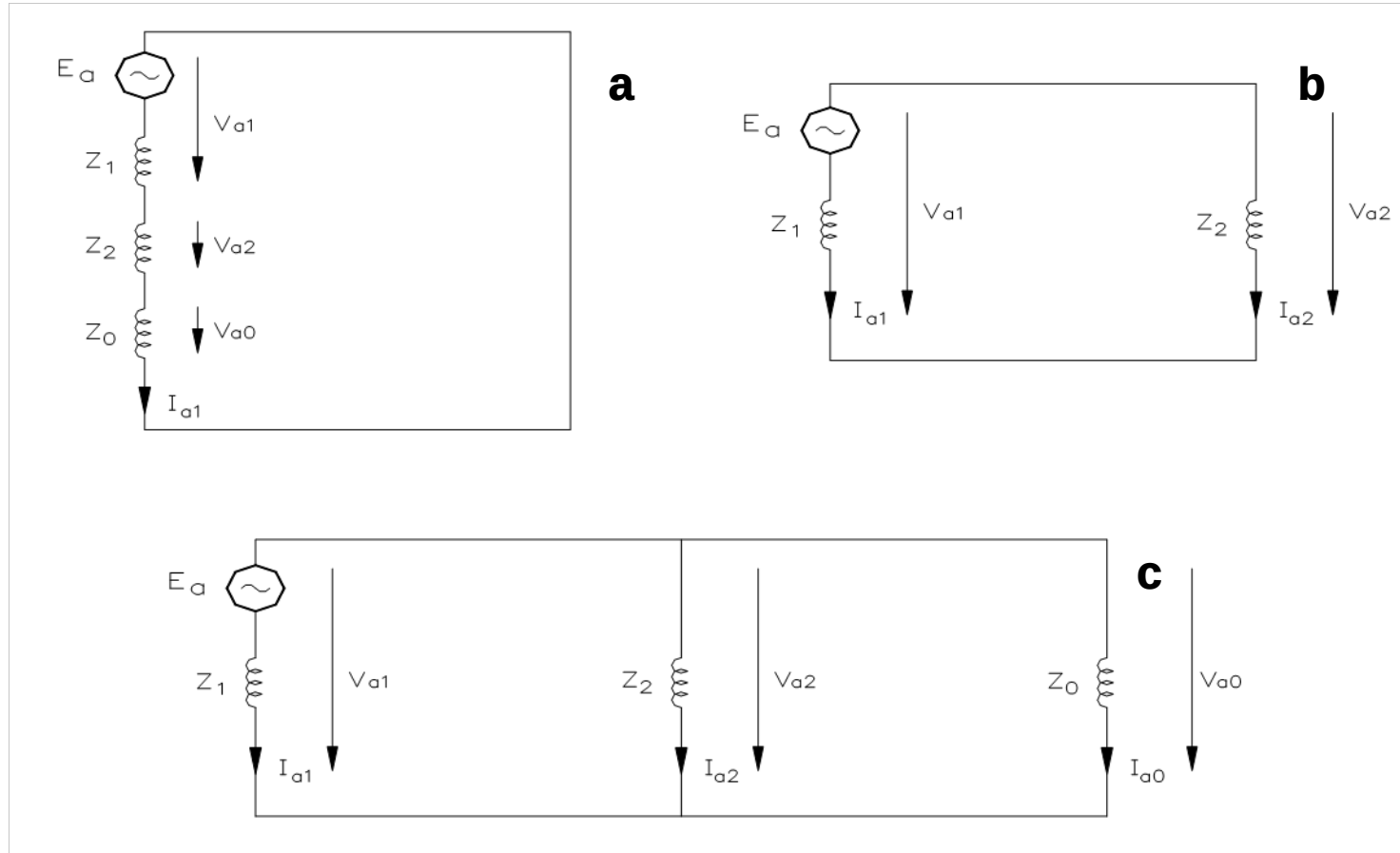
TRANSFORMADORES:

$$Z_0 = Z_1 = Z_2$$

GENERADORES Y MOTORES:

$$Z_0 \neq Z_1 \neq Z_2$$

CONEXIÓN DE REDES DE SECUENCIA PARA FALLAS ASIMÉTRICAS



- a Falla Fase a tierra
 - b Falla fase a fase
 - c Falla doble fase a tierra
-

IMPEDANCIAS DE SECUENCIA CERO EN TRANSFORMADORES DE POTENCIA

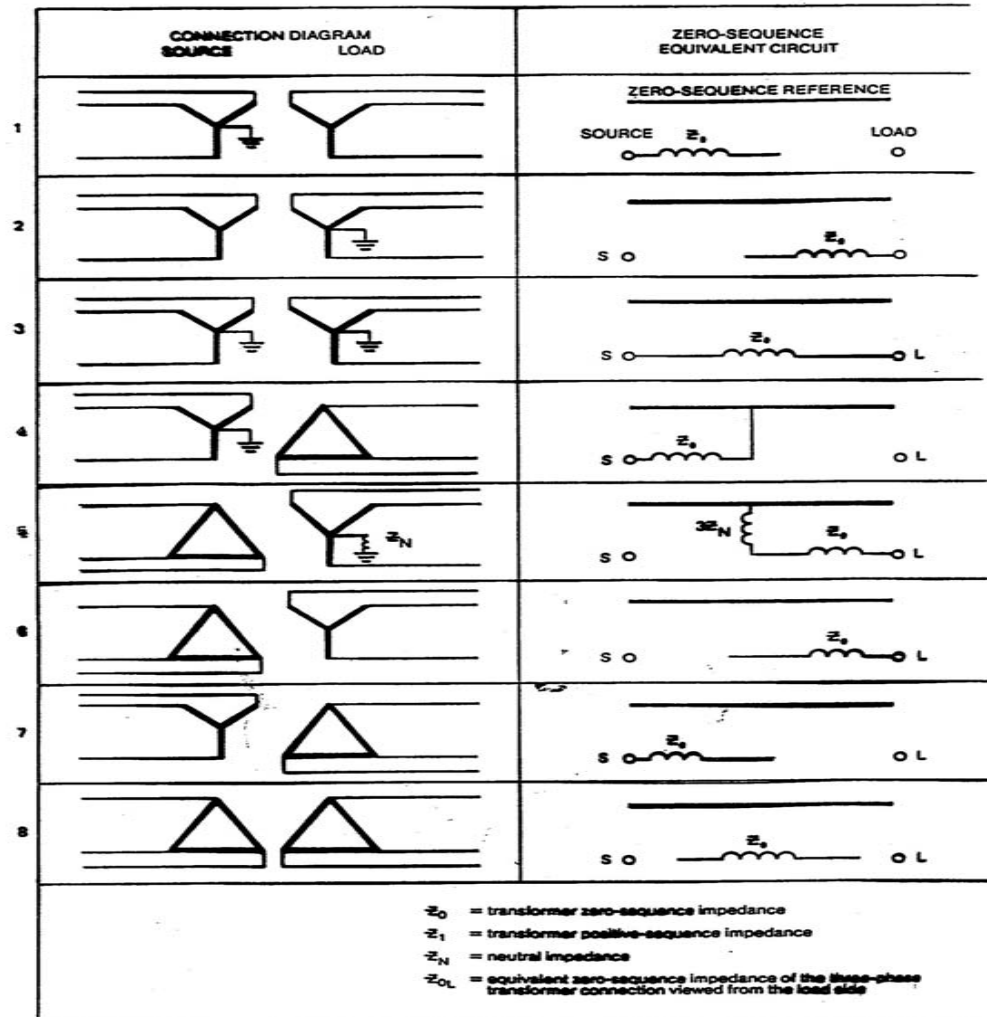


Figure 10A1.
Transformer connections and zero-sequence equivalent circuits.

TIPOS DE FALLA

Trifásica :

$$I_3 = \frac{V_{l-n}}{Z_{th\ e\ v.}}$$

Monofásica

$$I_1 = \frac{3 * V_{l-n}}{Z_0 + Z_1 + Z_2}$$

Fase-Fase

$$I_2 = \frac{\sqrt{3} V_{l-n}}{Z_1 + Z_2}$$

Fase-Fase-a tierra.

$$I_{2G} = \sqrt{3} V_{l-n} \frac{Z_0 - a^2 Z_2}{Z_1 Z_2 + (Z_1 + Z_2) Z_0}$$

CORRIENTES Y TENSIONES PARA VARIOS TIPOS DE FALLA

a

FAULT	POSITIVE SEQUENCE NETWORK CURRENT	NEGATIVE SEQUENCE NETWORK CURRENT	ZERO SEQUENCE NETWORK CURRENT	FAULT CURRENTS
a,b,c				
a,b				
b,c				
c,a				
a,b,e				
b,c,e				
c,a,e				
a,e				
b,e				
c,e				

a Corrientes de secuencia para diferentes tipos de falla
b Tensiones de secuencia para diferentes tipos de falla

b

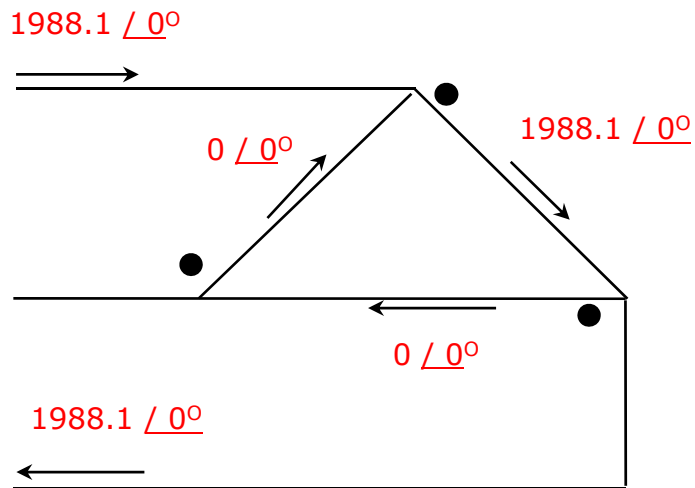
FAULT	POSITIVE SEQUENCE NETWORK VOLTAGE	NEGATIVE SEQUENCE NETWORK VOLTAGE	ZERO SEQUENCE NETWORK VOLTAGE	FAULT VOLTAGES
a,b,c				Zero a_1 Fault
a,b				
b,c				
c,a				
a,b,e				
b,c,e				
c,a,e				
a,e				
b,e				
c,e				

CIRCULACIÓN DE CORRIENTES DE FALLA

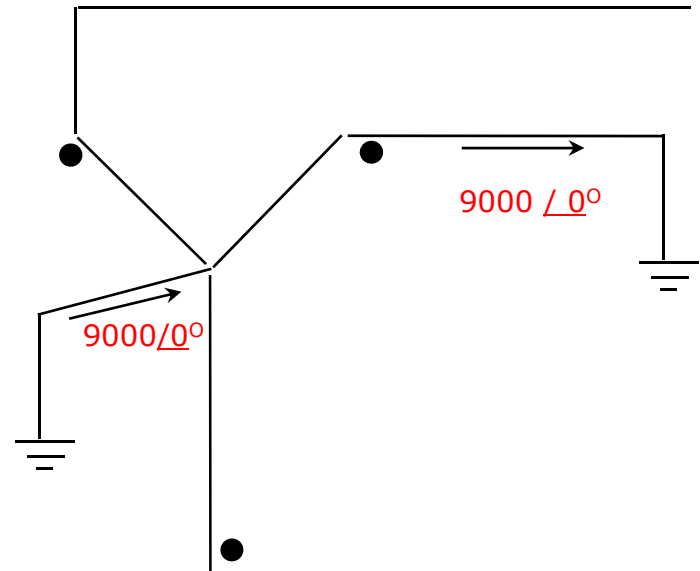
- ❏ Falla monofásica transformador delta-estrella
 - ❏ Falla fase - fase transformador delta-estrella
 - ❏ Falla monofásica transformador tridevanado estrella-estrella-delta
-

FALLA MONOFÁSICA

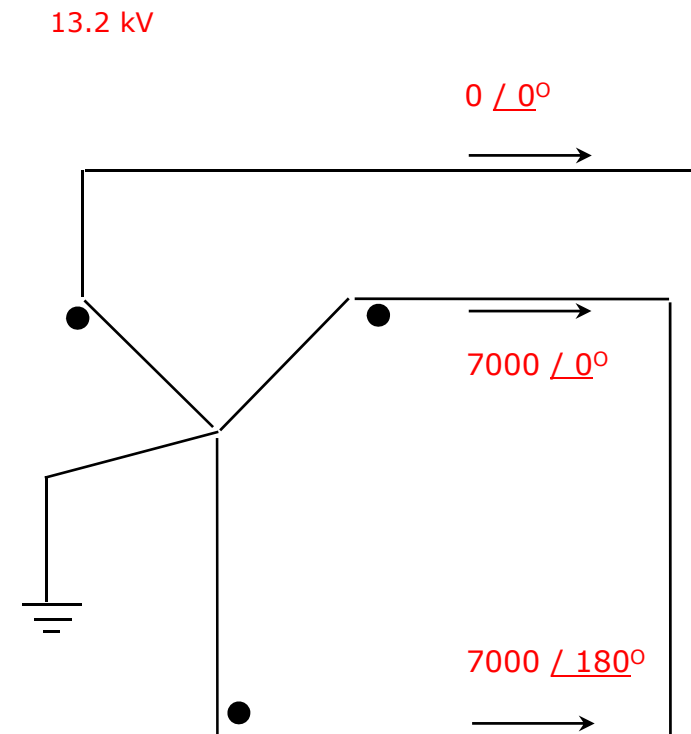
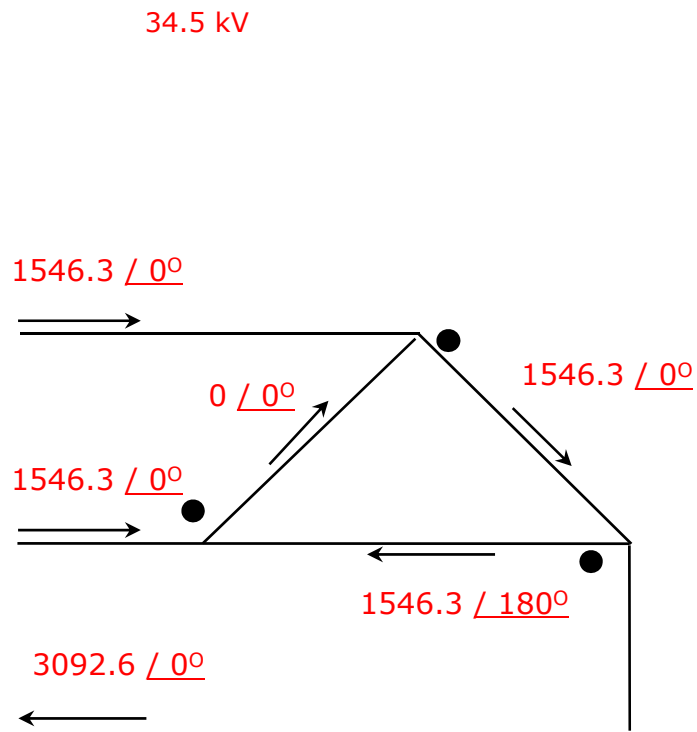
34.5 kV



13.2 kV



FALLA FASE - FASE



ESTUDIOS DE CORTOCIRCUITO

Baja tensión ($< 1 \text{ kV}$)

Corrientes de primer ciclo : Momentary duties.

Media (1 a 52kV) y Alta tensión (52 a 380kV)

Corrientes de primer ciclo:

Close & Latching Currents ($I_{c\&l}$) antes Momentary Current (I_{peak}), ANSI.

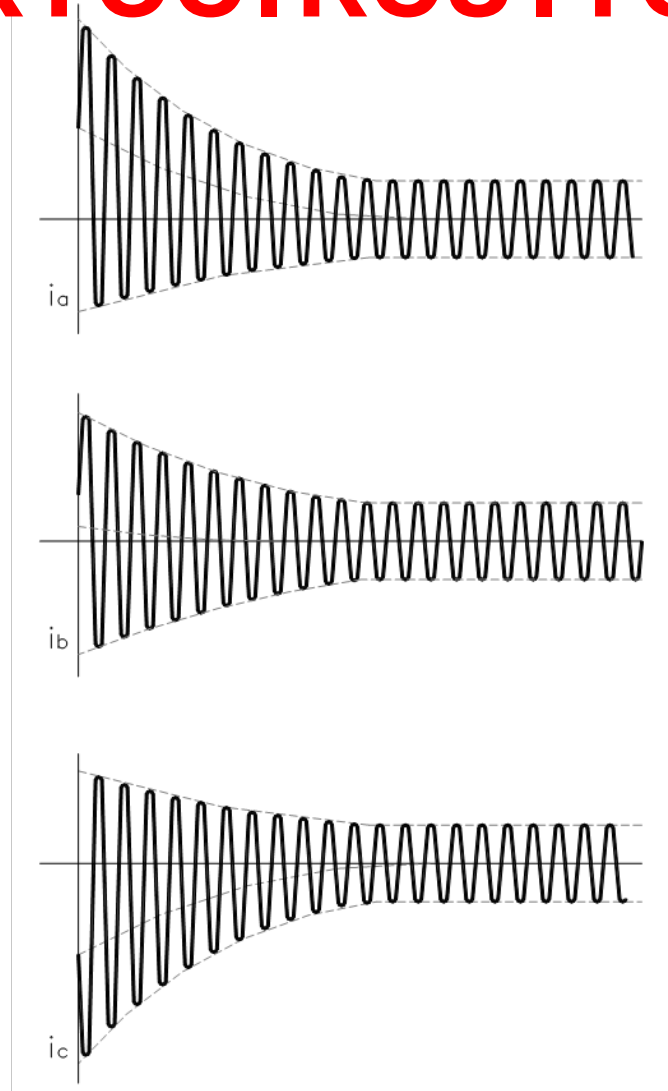
Making current (I_p), IEC.

Corrientes de interrupción:

Interrupting duties, ANSI.

Breaking current (I_b), IEC.

CORRIENTES DE CORTOCIRCUITO



ANÁLISIS DE CORTOCIRCUITO

- ☐ Determinación del nivel de cortocircuito en cada barraje del sistema eléctrico.
 - ☐ Determinación de los aportes de corriente de cortocircuito, por los elementos.
 - ☐ Determinación del valor $3I_o$, para fallas monofásicas
 - ☐ Los resultados tienen aplicación en la coordinación de protecciones
-

ANÁLISIS DE CORTOCIRCUITO

 NORMA DE CÁLCULO ANSI/IEEE C37.10

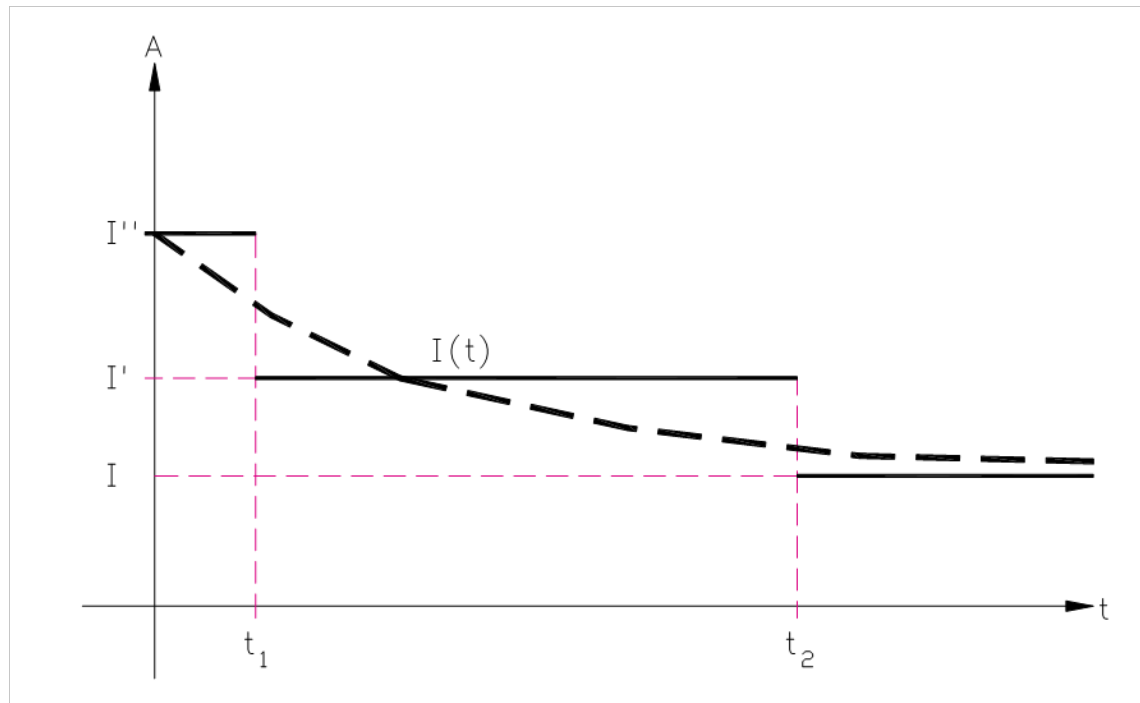
Corrientes de cortocircuito obtenidas

I_k'' : Momentánea o de primer ciclo

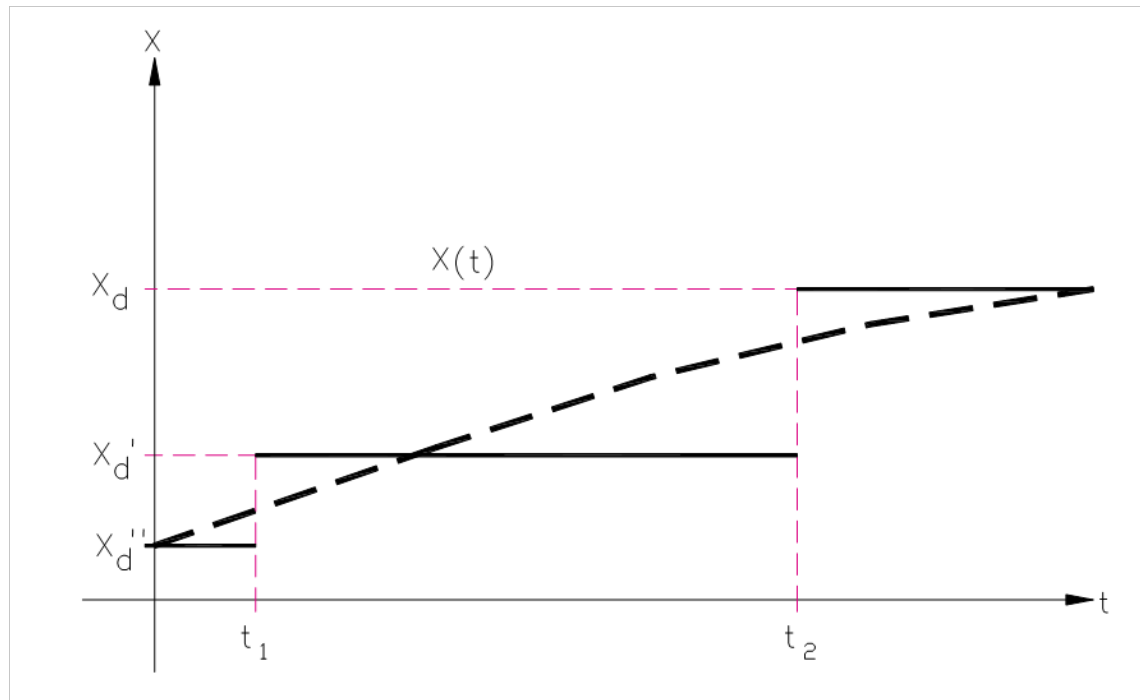
I_b : Interrupción

$3I_o$: Valor de 3 veces la corriente de secuencia cero

VARIACIÓN EN EL TIEMPO DE LA CORRIENTE DURANTE UNA FALLA



VARIACIÓN EN EL TIEMPO DE LA REACTANCIA DE UN GENERADOR DURANTE UNA FALLA



SIMULACIÓN DE GENERADORES

- ☐ Sólo interesa conocer la reactancia subtransitoria X'' . Valores típicos ver p. Ej.: Electrical Transmission and Distribution Reference Book (ETDRB) de Westinghouse.
 - ☐ En caso de no conocer la resistencia, IEC recomienda utilizar $R = 0.05 * X''$.
 - ☐ Para X_o , R_o , X_2 y R_2 , necesariamente datos del fabricante, o ETDRB Westinghouse.
-

SIMULACIÓN DE MOTORES

Durante un corto aportan corriente, de igual forma que los generadores, pero con las siguientes reactancias:

	C&L	INT
>1000 hp, 1800 r.p.m o menos	1.0 X"	1.5 X"
>250 hp, a 3600 r.p.m	1.0 X"	1.5 X"
Los no incluidos previamente		
mayores de 50 hp	1.2 X"	3.0 X"
Menores de 50 hp	1.67 X"	-----

SIMULACIÓN DE MOTORES

- ☐ Ante ausencia de información la IEEE recomienda utilizar $X'' = 0.16666$ para motores.
 - ☐ Para la R recomienda utilizar una gráfica.
 - ☐ Relación X/R .
-

CABLES

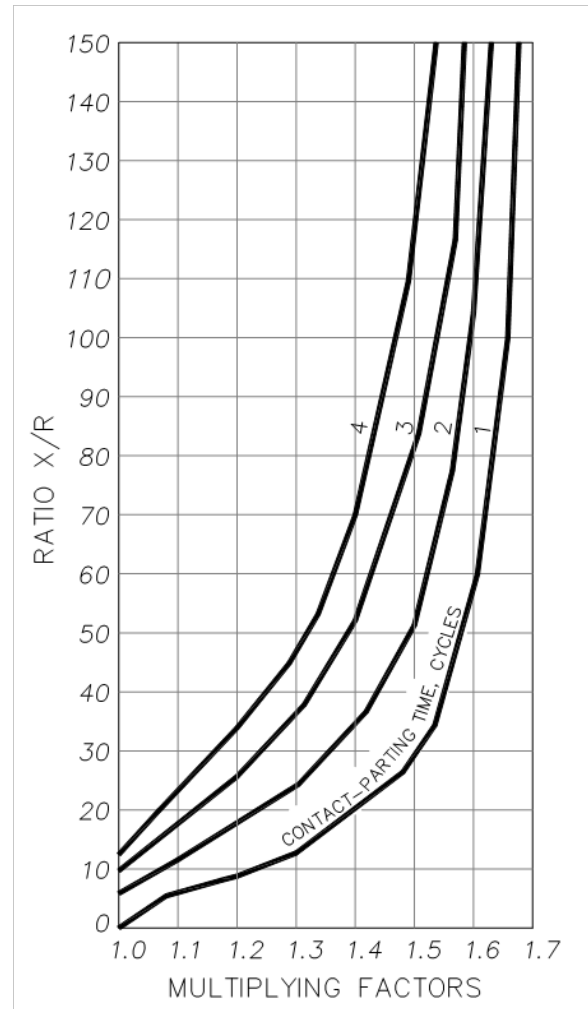
- ☐ Cálculo de impedancia de secuencia positiva
 - ☐ Impedancia de secuencia cero. Puesta a tierra de la chaqueta.
 - ☐ Longitud.
-

FACTOR MULTIPLICADOR

- ☐ Relación X/R
 - ☐ Trifásica o monofásica
 - ☐ Número de ciclos del interruptor.
 - ☐ Tipo de interruptor en baja y factores
-

FACTORES MULTIPLICADORES PARA FALLAS TRIFÁSICAS Y MONOFÁSICAS

IEEE STANDARD C37.5-1979



NEC 2005

Requieren marca de corriente de cortocircuito nominal:

Componentes

Control de Motores

Medidores

Montaje

Paneles Control Industrial

Paneles Eléctricos Maquinaria Industrial

Paneles HVAC (Heating, Ventilating, and Air Conditioning) no-residenciales de más de 60A



POR QUÉ????



Short Circuit Current Ratings (SCCR)

Qué es **Short Circuit Current Rating**?

- ✍ La máxima corriente de corto circuito que un componente, gabinete o equipo puede soportar seguramente cuando es protegido por un dispositivo específico de sobrecorriente o para un intervalo de tiempo específico
 - ✍ SCCR hace referencia a protección de componentes, gabinetes o paneles de control
-

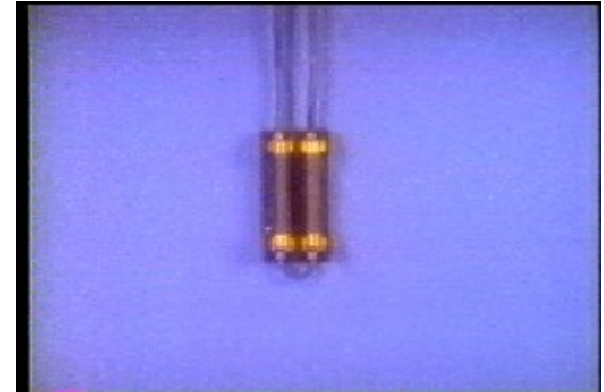
Short Circuit Current Ratings (SCCR)

Short Circuit Current Rating no es lo mismo que **Interrupting Rating**:

✍ **Interrupting Rating** es la máxima corriente que un fusible o interruptor pueden abrir de manera segura bajo condiciones de prueba normales

✍ **Interrupting Rating** sólo hace referencia a dispositivos de protección de sobrecorriente

✍ Adecuados **Interrupting Ratings** no aseguran protección de componentes de circuitos, gabinetes o equipos



Fusibles Clase H 10kA IR



Falla de 50,000A

Inadecuado **Interrupting Rating**

Marcación de rangos de corriente de cortocircuito

Por qué es importante la marcación de rangos de corriente de cortocircuito?

- ☐ Necesidad de asegurar cumplimiento de NEC® 110.10
 - ☐ Ayuda a eliminar riesgos donde componentes y equipos están empleados por encima de sus rangos
 - ☐ Simplifica proceso de inspección para aprobación
-

Marcación de rangos de corriente de cortocircuito

Component Marking Requirements

- ❏ Short Circuit Current Rating may be established during testing as part of the listing and labeling process for individual components or multiple component assemblies
-

Marcación de rangos de corriente de cortocircuito

230.82(3) – Equipment Connected to the Supply Side of Service Disconnect.

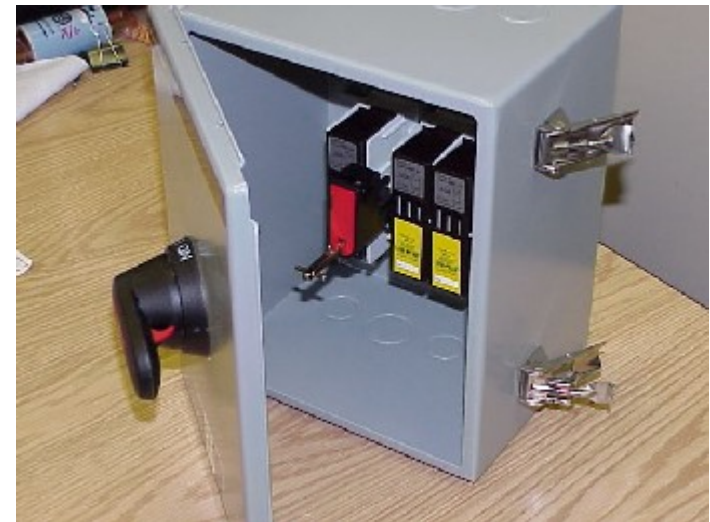
Only the following equipment shall be permitted to be connected to the supply side of the service disconnecting means:

(3) Meter disconnect switches nominally rated not in excess of 600 volts that have a **short-circuit current rating equal to or greater than the available short circuit current**, provided all metal housings and service enclosures are grounded.

Marcación de rangos de corriente de cortocircuito

230.82(3) Meter Disconnect Switches:

- ☐ Must have a marked short circuit current rating equal to or greater than the available short circuit currents
- ☐ Typically achieved by fused disconnect utilizing current-limiting fuses



Marcación de rangos de corriente de cortocircuito

430.8 – Motor Controllers

A controller shall be **marked** with the manufacturer's name or identification, the voltage, the current or horsepower rating, **the short-circuit current rating**, and such other necessary data to properly indicate the applications for which it is suitable.



Marcación de rangos de corriente de cortocircuito

430.8 Motor Controllers

Exceptions where the Short Circuit Current Rating is **not** required on the controller:

- 📖 1/8HP or less motors which are normally left running and constructed not to be damaged by overloads
 - 📖 1/3HP or less portable motors where the controller is the attachment plug and receptacle
 - 📖 The rating is marked elsewhere on an assembly
 - 📖 The assembly into which the controller is to be installed is marked with a rating
 - 📖 Controller is rated 2HP or less at 300V or less and is listed exclusively for general purpose branch circuits
-

Marcación de rangos de corriente de cortocircuito

Control de motores

- UL 508 has:
 - 📖 "Standard" fault current test
 - 📖 An optional "high available" fault current test
 - 📖 Optional Type 2 "no damage", "high available" fault current (UL 508E)
 - 📖 "Standard" level:
 - 5kA for 0 - 50HP ratings
 - 10kA for 51 - 200HP ratings, etc.
 - Current limiting fuses are often used in the optional "high-available" fault current tests and Type 2 "no damage" tests to achieve high short circuit current ratings
-

Marcación de rangos de corriente de cortocircuito



GENERAL PURPOSE SWITCH
INTERRUPTEUR, USAGE GENERAL

Short circuit rating 100kA at 600VAC max
when protected by 100A class J or T
5kA when protected by 150A class H or
RK5 fuses

LISTED 3E73
MAN MTR CNTRL



Motor Controller Label Example
(from an 80A, 40HP rated controller)

Marcación de rangos de corriente de cortocircuito

Equipment Marking Requirements

-  Short Circuit Current Rating can be established during testing as part of the Listing and Labeling process
 -  Where testing is not feasible, Short Circuit Current Ratings can be determined using approved engineering methods
-

Marcación de rangos de corriente de cortocircuito

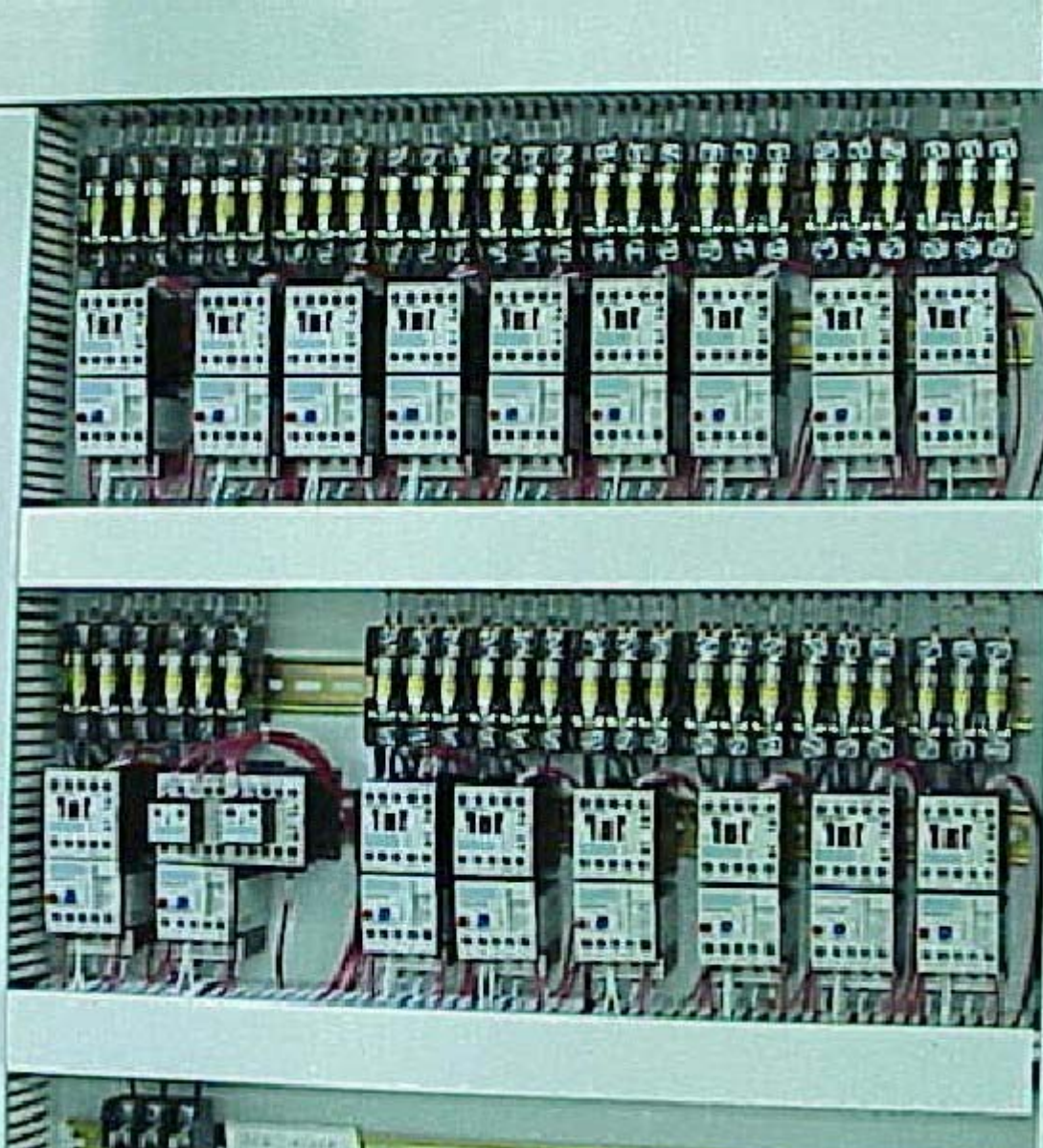
409.110 – Industrial Control Panels – Marking.

An industrial control panel **shall be marked** with the following information that is plainly visible after installation:

(3) **Short-circuit current rating** of the industrial control panel based on one of the following:

- a. Short-circuit current rating of a listed and labeled assembly
- b. Short-circuit current rating established utilizing an approved method

FPN: UL 508A-2001, Supplement SB, is an example of an approved method



Industrial Control Panels: Now Marked with Short Circuit Current Rating

Marcación de rangos de corriente de cortocircuito

670.3 – Industrial Machine Nameplate Data.

(A) **Permanent Nameplate.** ... shall be attached to the control equipment enclosure or machine and shall be plainly visible after installation. The **nameplate shall include** the following information:

(4) **Short-circuit current rating** of the industrial control panel based on one of the following:

- a. Short-circuit current rating of a listed and labeled assembly
- b. Short-circuit current rating established utilizing an approved method

FPN: UL 508A-2001, Supplement SB, is an example of an approved method

Marcación de rangos de corriente de cortocircuito

Plastics Processing Machine	
Serial Number	SN2356YUP77
Current	87 Amperes
Largest Motor H.P.	25 Horsepower
Max OCP Device	60 Ampere
Voltage	460 - 480 volts
Phase & Freq..	3ph., 60 Hz
Short Circuit Current Rating 100,000 Amperes RMS	
Diagram Numbers	CM 12.1 THRU CM 12.5
Quality Machine Tool Somewhere, USA	

~~Example: Industrial Machinery Control Panel Label~~

Marcación de rangos de corriente de cortocircuito

440.4(B) – Marking on Hermetic Refrigerant Motor-Compressors and Equipment

(B) Multimotor and Combination-Load Equipment.

Multimotor and combination-load equipment shall be provided with a visible **nameplate marked with** the maker's name, the rating in volts, frequency and number of phases, minimum supply circuit conductor ampacity, the maximum rating of the branch-circuit short-circuit and ground-fault protective device, and **the short-circuit current rating of the motor controllers or industrial control panel.**

Marcación de rangos de corriente de cortocircuito

Combination Load and Multimotor HVAC and Refrigeration Equipment

Exceptions:

- 📖 Equipment used in one and two family dwellings
 - 📖 Cord-and-attachment-plug connected equipment
 - 📖 Equipment supplied by a branch circuit protected at 60A or less
-

Marcación de rangos de corriente de cortocircuito

HVAC Control Panel	
Serial Number	HVDB708429521
Current	72 Amperes
Min Circuit Ampacity	90 Amperes
Max Fuse Size	125 Ampere
Voltage	460 - 480 volts
Phase & Freq..	3ph., 60 Hz
Short Circuit Current Rating 40,000 Amperes RMS	
HVAC Equipment, Inc. Anytown, USA	

Ejemplo de etiqueta para **HVAC**

Asegurar cumplimiento

For equipment requiring Marked Short Circuit Current Ratings

- Engineer provides:
 - 📖 Available short circuit currents at each installation point
 - 📖 Short circuit current rating of each piece of equipment or panel
 - During site inspection, inspector compares actual marked short circuit current ratings to the submitted data: planned SCCR and available short circuit currents
-

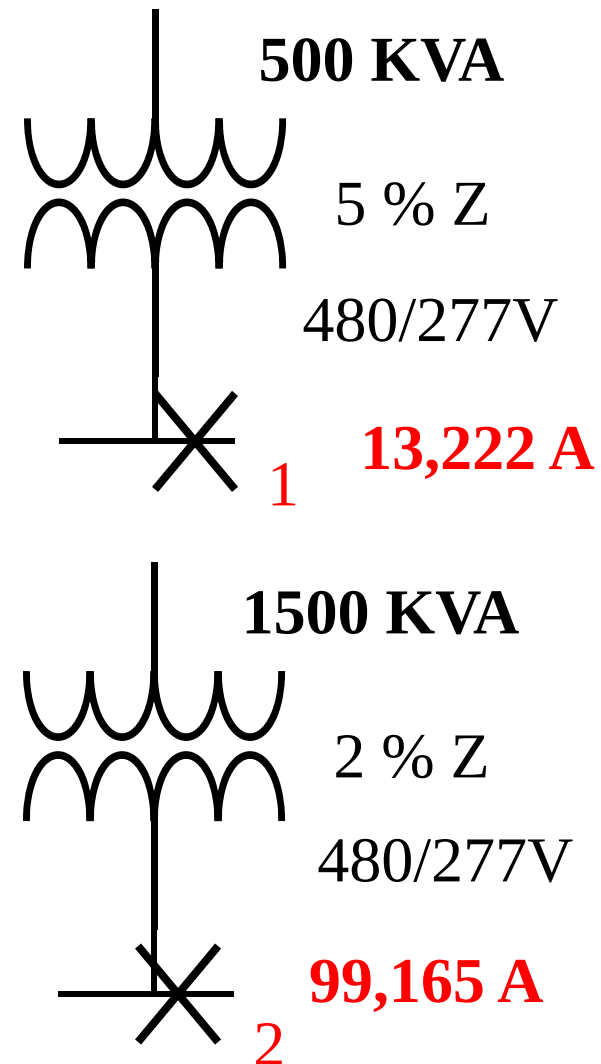
Asegurar cumplimiento

Este método requiere ingeniería y análisis por ingenieros de diseño y revisión adecuada de inspectores.



Ensuring Compliance: Simple Check For Short Circuit Current Rating

- Determine the maximum, worst case short circuit current available at the terminals of the supply transformer
- Verify that all required equipment is marked with a short circuit current rating sufficient for this maximum, worst case available current
- If SCCRs are sufficient: installation approved. If this SCCRs insufficient by this quick check method, a detailed analysis may be required



Achieving High Short Circuit Current Ratings

High Short Circuit Current Ratings Make Equipment and Controllers:

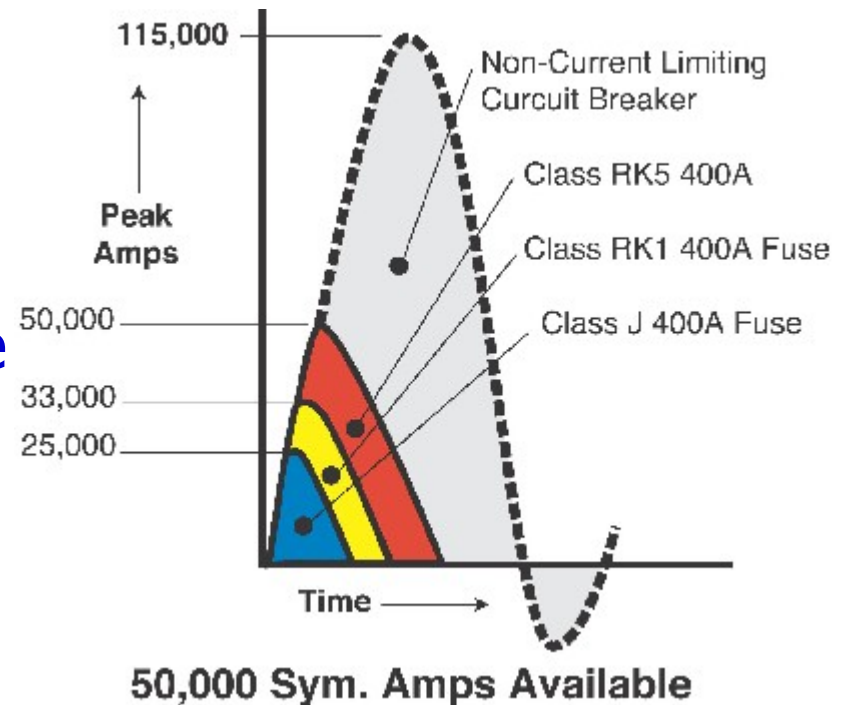
- Easier to specify and install for compliance
- More flexible – can be moved from location to location safely



Alcanzan rangos altos de corriente de corto circuito

Fusibles limitadores de corriente:

- Reducen energía de la falla
- Pueden emplearse para alcanzar rangos altos de corriente de corto circuito en controles de motores, ensambles de múltiples componentes, desconectadores y paneles de control industriales.



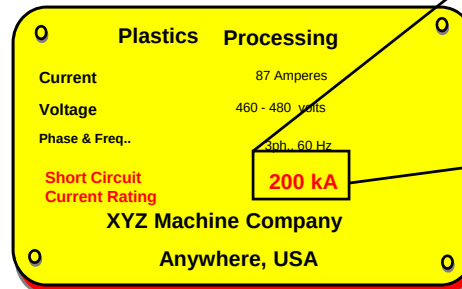
Regulatory - 2005 NEC® Changes

Marked Short Circuit Current Ratings

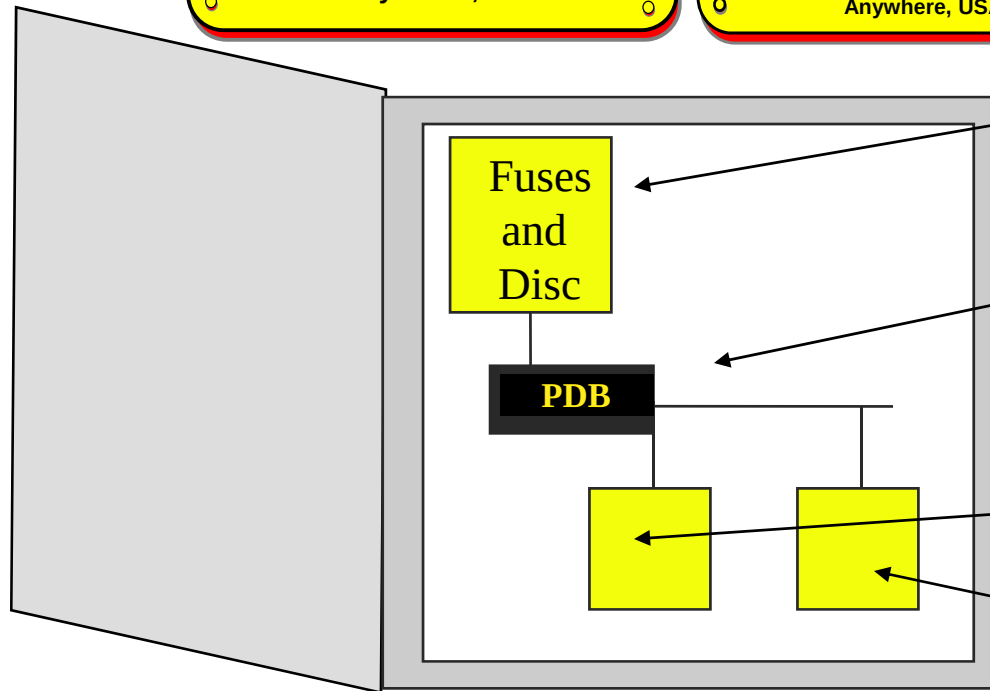
Before



Now Marked



200 kA SCCR



400A Class J Fuse Disconnect
Listed 200,000A SCCR

Power Distribution Block Listed
200,000A SCCR Protected by
400A Class J Fuses

Branch circuits with current
limiting fuses, contactors and
overloads
Listed 200,000A SCCR

Marked Short Circuit Current Ratings

Summary: The 2005 NEC® now requires short circuit current ratings to be marked on:

- Meter Disconnect Switches
 - Motor Controllers
 - Industrial Control Panels
 - Industrial Control Panels for Industrial Machinery
 - Combination Load and Multimotor HVAC and Refrigeration Equipment
-

